

Comp 442 Group Assignment 1

“Phish Detector”

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Revision History

Version	Date	Author(s)	Notes
v1.0	6/6	Haley	Google doc created and outline written for members
v1.1	6/6	Full team	Section 1 written, section 2 started
v1.2	6/7	Full team	Section 2 and 3 written
v1.3	6/8	Haley	Full review of doc. Prepared for submission

1. Dataset Acquisition

1.1 Dataset Information

- Dataset Name: Email Spam Detection Dataset
- Source: Kaggle
- Location: <https://www.kaggle.com/datasets/abdmental01/email-spam-dedection>

1.2 Method of Acquisition

The dataset was downloaded from Kaggle and will be imported into the project environment for preprocessing, analysis, and model development after review is complete.

1.3 Problems Encountered

- Initial review was required to verify dataset structure and labeling consistency.
- Potential issues such as missing values, duplicate records, and class imbalance may exist and will be investigated during preprocessing.

1.4 Resolutions

- Data cleaning techniques will be applied, including duplicate removal, missing value handling, and label verification.
- Class distribution will be analyzed and appropriate balancing techniques will be considered if necessary.

2. Dataset Description

2.1 Dataset Format

The dataset is stored as a CSV file containing labeled email messages for spam detection. Each record represents a single email message and its corresponding classification label.

2.2 Dataset Quantity

The dataset contains 5572 records and 2 attributes

2.3 Dataset Fields

Message: The content of the email message.

Category: The classification label indicating whether the message is Ham or Spam.

2.4 Data Characteristics

The dataset consists of email messages labeled as either spam or ham. The labels are used as the target variable for supervised machine learning. The message field contains the text content that will be analyzed and transformed into numerical features for identification.

2.5 Data Structure

The dataset is stored in a structured tabular format because it is organized into rows and columns within a CSV file. However, the Message field contains unstructured text data, which requires NLP techniques before it can be used by machine learning algorithms.

2.6 Dataset Type

The dataset can be classified as tabular data and text data

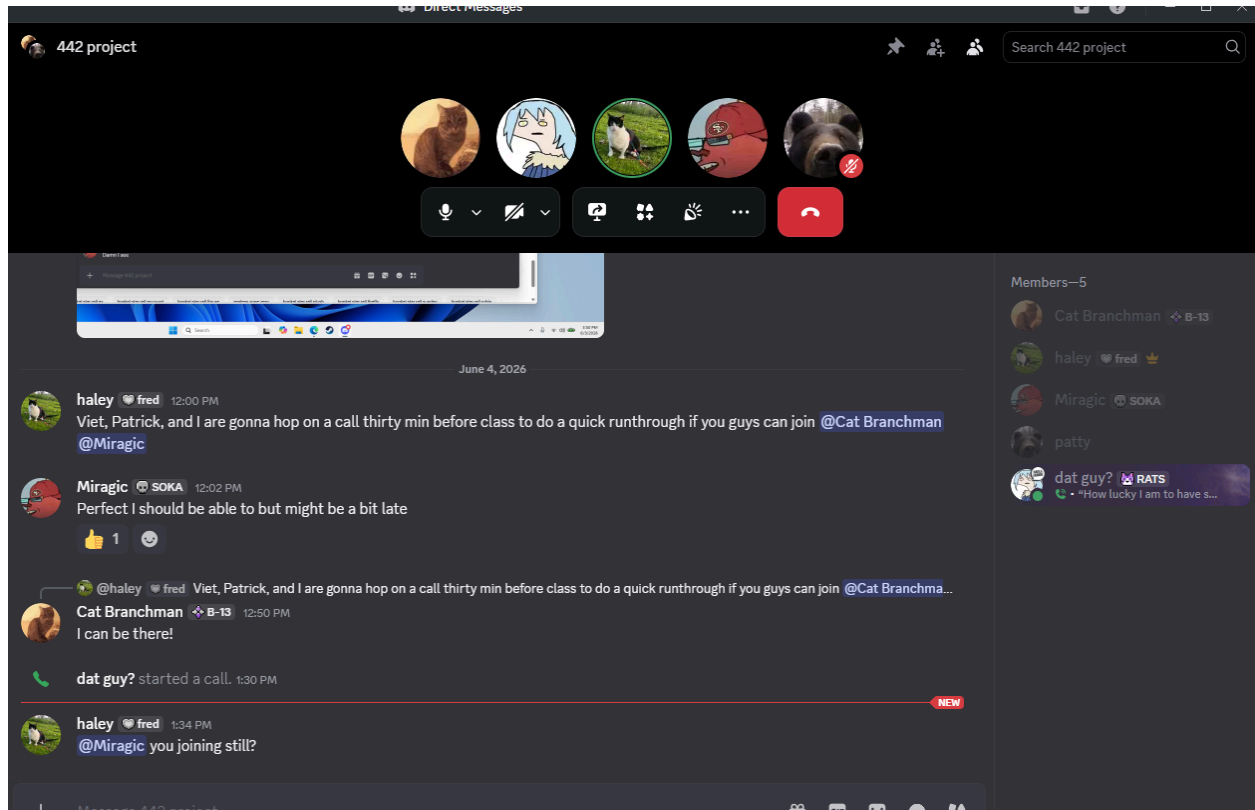
2.7 Feature (Attribute) Types

The category attribute is categorical feature, nominal feature, binary feature. Possible values are Ham and Spam. The Message attribute is text feature and unstructured feature. Additional features may be created during preprocessing, including message length, word count, number of URLs, TF-IDF values, and the number of capitalized words.

3. Text Feature Extraction

This project uses text data in the form of email messages. Since machine learning algorithms can't directly process raw text, NLP techniques will be used to extract meaningful features from the email content and convert them into numerical representations. Before feature extraction, the email messages will be put through preprocessing steps including converting text to lowercase, removing punctuation and special characters, tokenization, stopword removal, and lemmatization or stemming. We will extract the following features from the email messages: TF-IDF, message length, word count, URL count, suspicious keyword frequency, and capitalization patterns. After extraction, the features will be transformed into numerical vectors that can be preprocessed by machine learning algorithms such as Naive Bayes, Logistic Regression, SVM, and Random Forest classifiers.

Appendix



Shows proof of collaboration of all team members via discord call.